

● SNOWFLAKE ELITE PARTNER

"That's just semantics."

Best practices when designing and deploying Snowflake Semantic Views.

Stéphane Burwash — Lead Customer Solutions Architect, Infostrux

Everything is Agents these days.

Large investment and re-direction of capital into **agentic design** and **LLM-based features**.

ROI remains unclear - Some initiatives have **incredible saving opportunities** of **are great revenue drivers**. Others are **money pits**, with high Research & Development and token costs for little yield.

BUT THEY ARE

Not running in production

Not accurate

Not tied in to the broader knowledge of the company.

Not scalable or predictable

People are looking to move from the POC ————— to the fully automated workload.

INTRODUCTION

Pie in the sky.

What does successful agent design look like?

WHAT WE'VE SEEN

80% Reduction

in ticket management, by offboarding small requests to your agent.

3 Full Time Equivalents

re-allocated to more impactful analytical work by enabling self-serve analytics — at \$400/month in token costs.

THE OBJECTIVE — FROM AN ARCHITECTURE STANDPOINT

Accurate

"This answer represents reality"

They perform the correct action based on your best practices.

Predictable

"Different users get the same information"

The language of an answer can change; the core of the answer stays predictable as the system iterates.

Trustworthy

"My systems remain secure"

Only relevant information, plus the audit trail required to ascertain responses.

Data system design in a nutshell.



AGENT DESIGN, PART ONE

Technical architecture

Models, orchestration, tools, plumbing.

AGENT DESIGN, PART TWO

Data architecture

What the agent is allowed to know, and what the words mean.

Data system design in a nutshell.

AGENT DESIGN, PART ONE

Technical architecture

Models, orchestration, tools, plumbing.

AGENT DESIGN, PART TWO

Data architecture

What the agent is allowed to know, and what the words mean. **Today's topic.**

01 What are the core components of an agentic workflow?

02 How do you create an end-to-end workflow?

03 How do you iterate on this workflow?

01

VOCABULARY

Context, semantics & consumption.

Three words we all use, and mostly don't mean the same thing by.

STORY

Le Petit Prince.

Asked for a sheep, the pilot draws a box and says the sheep is inside. The prince is satisfied.

Asked for what good governance and agentic design look like, everyone has an opinion on what it is **not** — and a bit more difficulty identifying what it **has**.

Let's identify what this sheep should look like.



*the sheep
is in here*

DEFINITIONS

Core concepts.

INTERPRETATION

Context

Provide operational, temporal, and governance context for reasoning.

ENFORCEMENT

Semantics

Standardize metric definitions for consistent analytics.

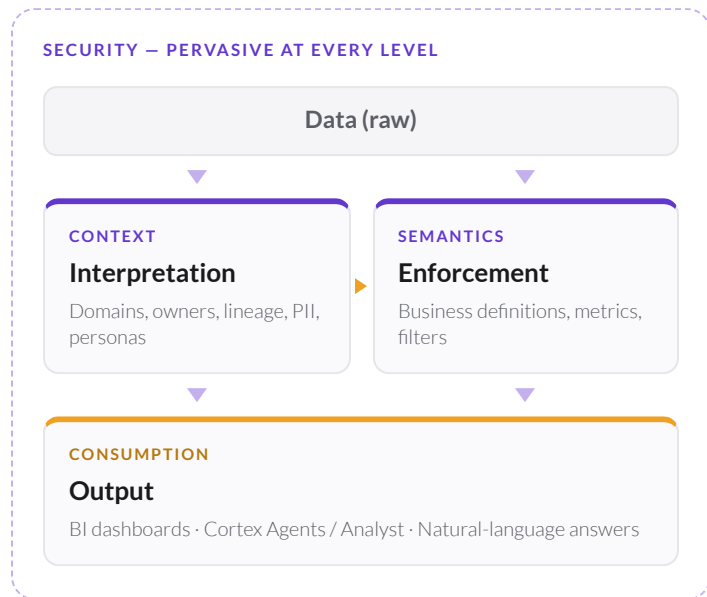
OUTPUT

Consumption

Use governed context and semantics to produce a trustworthy answer, at the point where a human or an agent actually acts on the data.

These concepts are tool-agnostic — and hard to tell apart, because one tool often spans several of them.

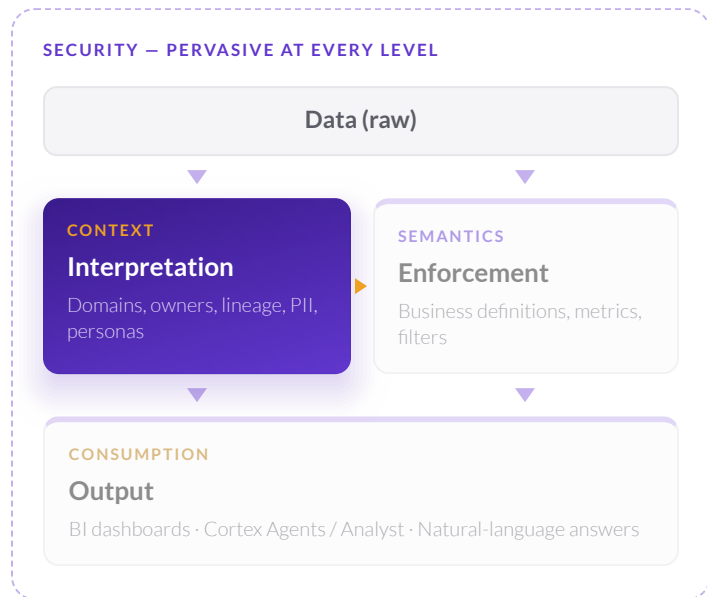
Architecture.



OVERVIEW

- ◆ Context and semantics **both feed into consumption.**
- ◆ Context provides the general knowledge.
- ◆ Semantics enforce business definitions.
- ◆ Output is consumed downstream.
- ◆ Security is **pervasive at every level.**

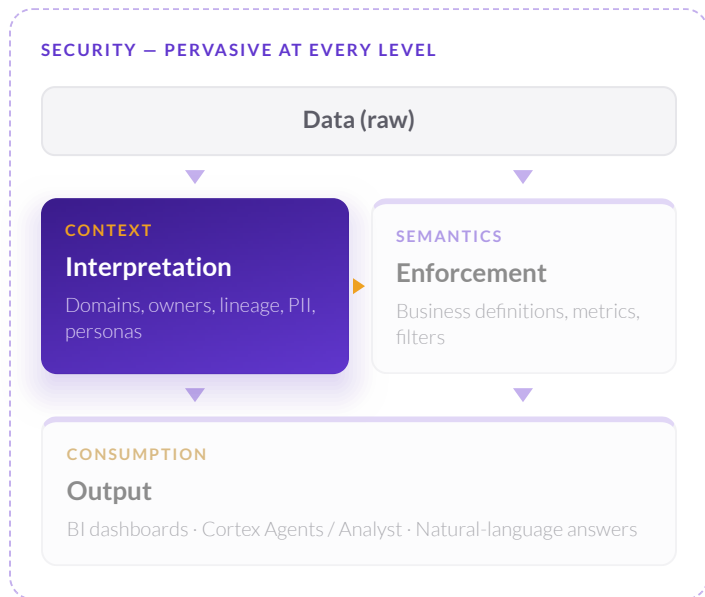
What is context?



OVERVIEW

- ◆ Data & metadata
- ◆ Ownership — stewards & data stewards
- ◆ Business process
- ◆ Ontology
- ◆ Segmentation & curation — data domains and data products

What is context?



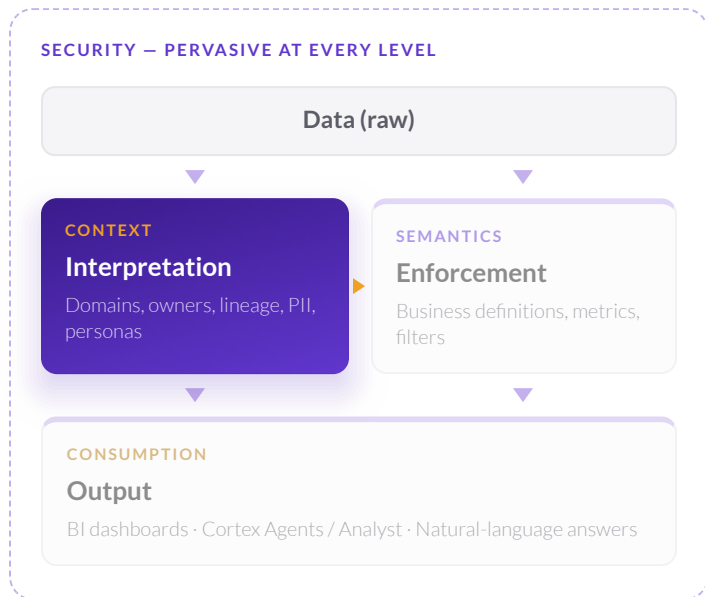
OVERVIEW

- ◆ Data & metadata
- ◆ Ownership — stewards & data stewards
- ◆ Business process
- ◆ Ontology
- ◆ Segmentation & curation — data domains and data products

WHERE IS IT?

- ◆ **Centralized data catalog**
- ◆ **Source tools** — Slack, Confluence, tickets: wherever knowledge actually gets written down.
- ◆ **Data platform** — comments, tags and object metadata, right next to the data.

What is context?



OVERVIEW

- ◆ Data & metadata
- ◆ Ownership — stewards & data stewards
- ◆ Business process
- ◆ Ontology
- ◆ Segmentation & curation — data domains and data products

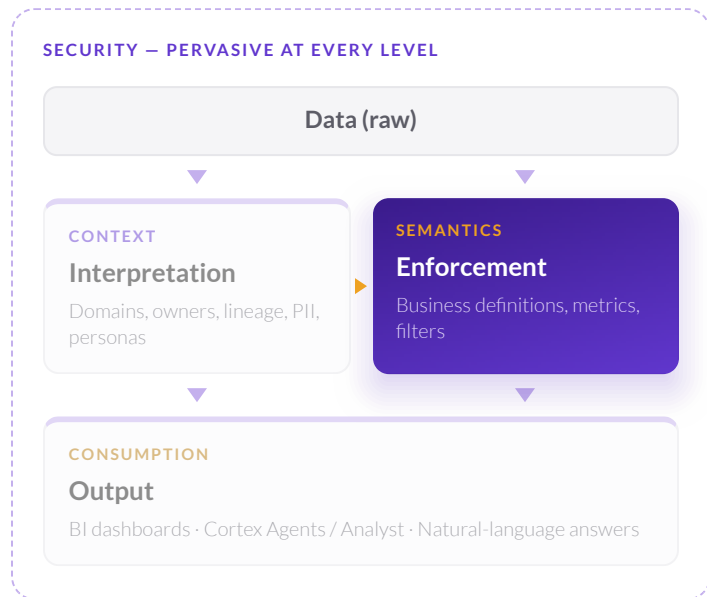
WHERE IS IT?

- ◆ Centralized data catalog
- ◆ Source tools — Slack, Confluence, tickets

PAIN POINTS

- ◆ **Identifying relevant information** — metadata, key stakeholders, and everything nobody wrote down.
- ◆ **Performing the modeling** — ontology, business process modeling: slow, human work.

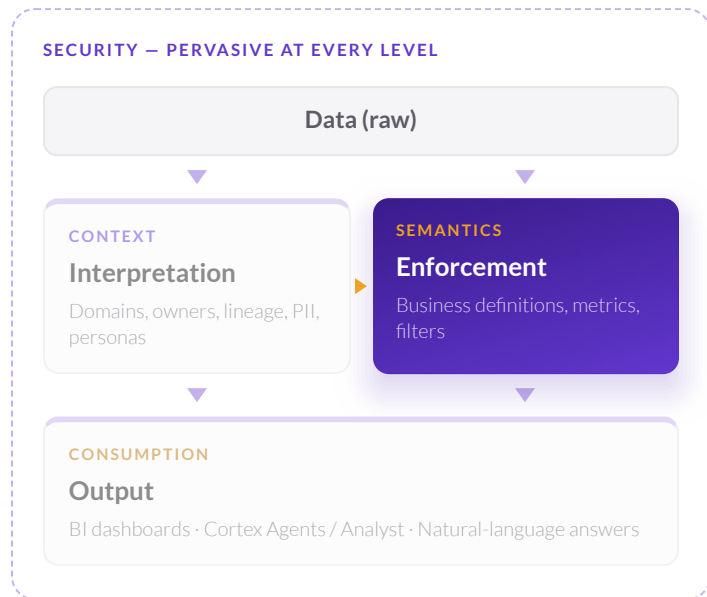
What are semantics?



OVERVIEW

- ◆ Business definitions
- ◆ Metric definitions
- ◆ Key filters
- ◆ **Re-useable & deterministic**

What are semantics?



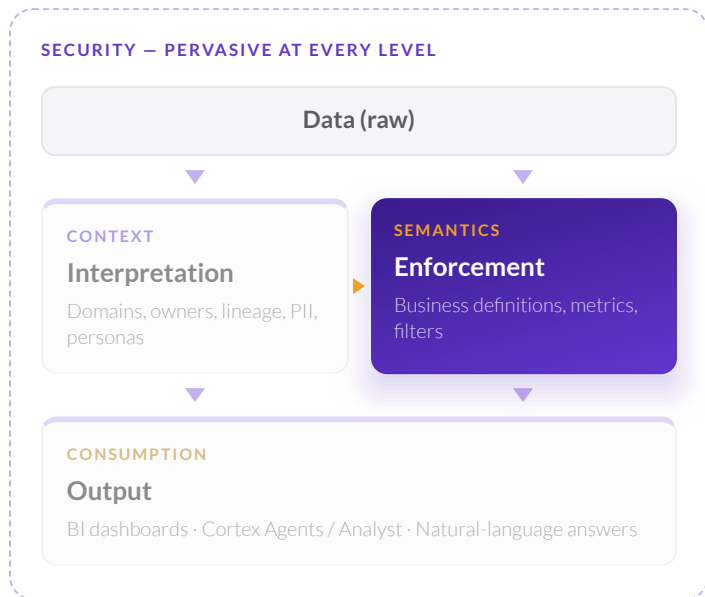
OVERVIEW

- ◆ Business definitions
- ◆ Metric definitions
- ◆ Key filters
- ◆ Re-useable & deterministic

WHERE IS IT?

- ◆ BI tools
- ◆ Data models
- ◆ Semantic definitions for agents

What are semantics?



OVERVIEW

- ◆ Business definitions
- ◆ Metric definitions
- ◆ Key filters
- ◆ Re-useable & deterministic

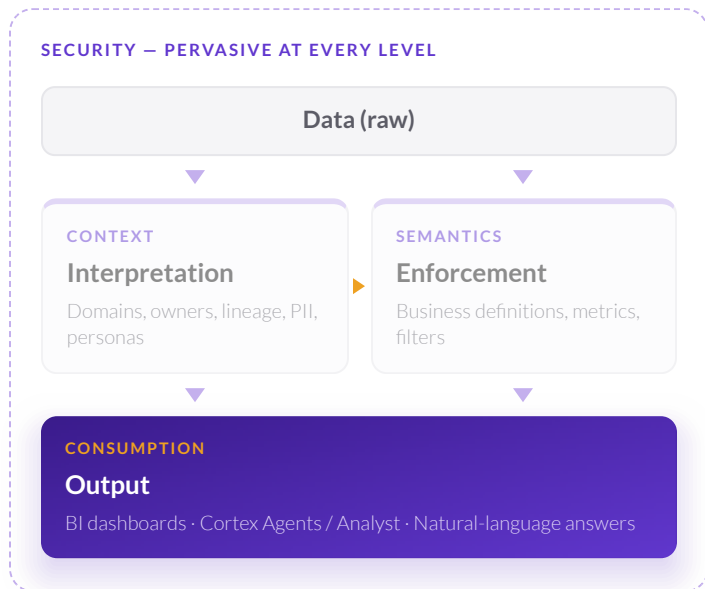
WHERE IS IT?

- ◆ BI tools
- ◆ Data models
- ◆ Semantic definitions for agents

PAIN POINTS

- ◆ Identifying proper definitions across different teams
- ◆ Keeping context and semantics in sync

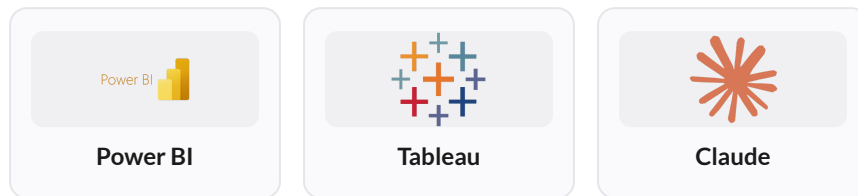
What is consumption?



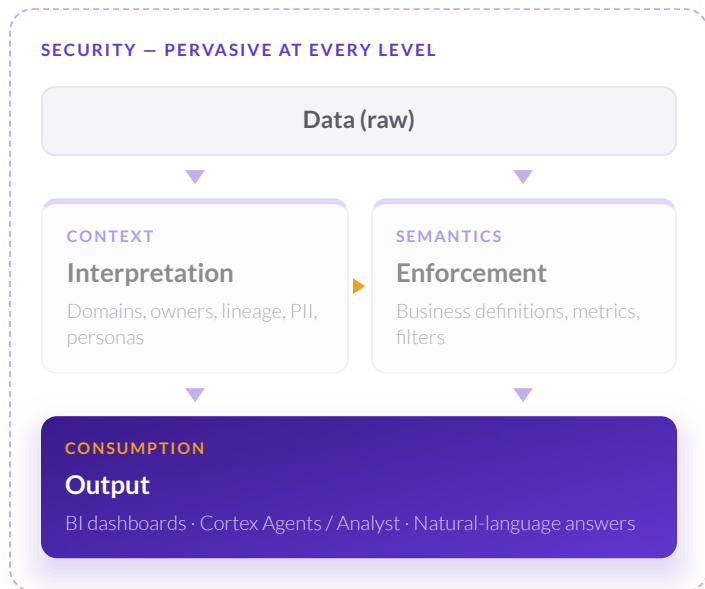
OVERVIEW

- ◆ How context and semantics are leveraged for downstream consumption – autonomous agents, analytics, dashboards.
- ◆ Historically only BI. Now the same requirement exists twice – once for humans, once for agents.

WHERE IT LANDS



What is consumption?



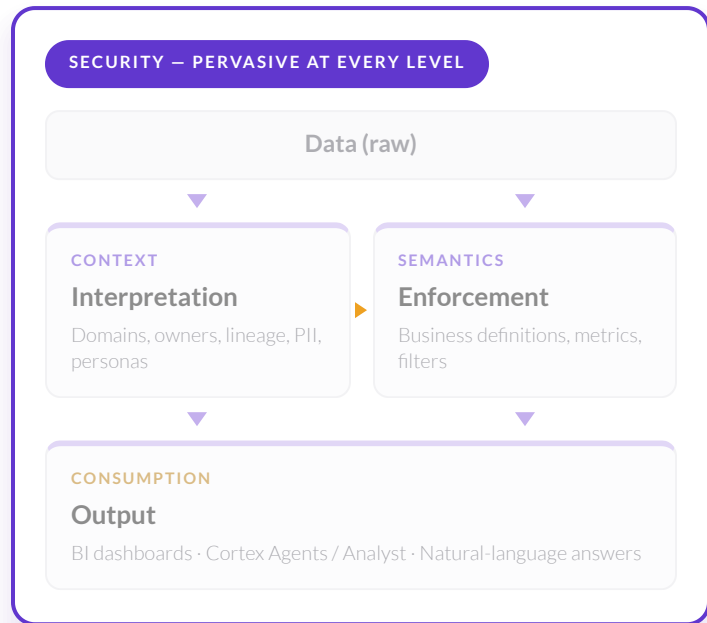
OVERVIEW

- ◆ Leveraged downstream — autonomous agents, analytics, dashboards.
- ◆ Historically only BI. Now the requirement exists twice — humans and agents.

PAIN POINTS

- ◆ **Duplication, sprawl & silos**
- ◆ **Upstream changes** with uncontrolled, unmonitored impacts downstream

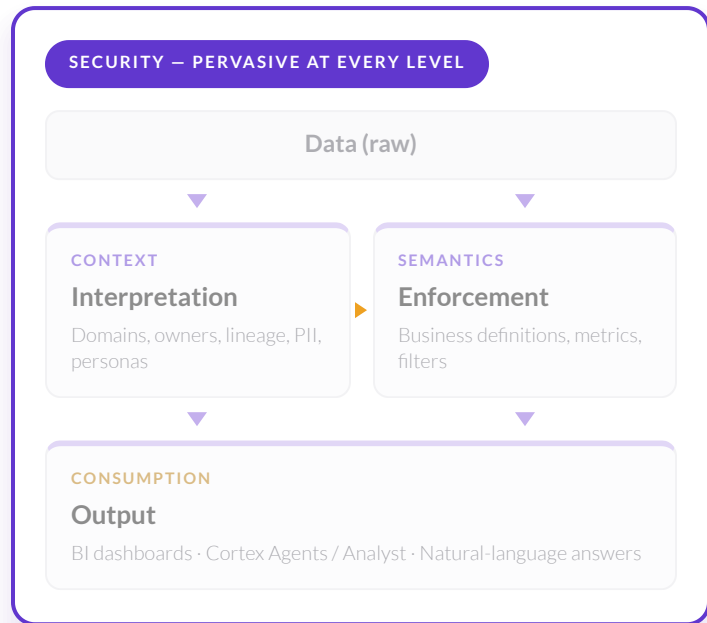
What is security?



OVERVIEW

- ◆ Enforcement of the policies declared within the context.
- ◆ **Data masking**
- ◆ **Data access**

What is security?



OVERVIEW

- ◆ Enforcement of the policies declared within the context.
- ◆ Data masking
- ◆ Data access

PAIN POINTS

- ◆ **How to tie enforcement back to policy**
- ◆ **How to avoid duplicating enforcement effort**

02

IMPLEMENTATION

How do you build out an
end-to-end workflow?

STORY

Le Petit Prince.

We know what the sheep should look like. A box with a promise inside is not a design.

Let's draw out this sheep.

*now we draw
what's inside*

Example

THE CLIENT

St-Exupé Corp

A consulting company giving advice on how to successfully fly and live through space.

RAW DATA

Flight

Time sheets, flight plans, plane models, geolocation of cities.



CONTEXT

Metadata

Held natively in Snowflake.



SEMANTICS

Data Models & Semantic Views

Orchestrated in dbt, materialized in Semantic Views



CONSUMPTION

Cortex Agents → Teams

Surfaced through MCP.

Business process modeling.

CONTEXT Column metadata (Snowflake)

SEMANTICS Snowflake Semantic Views

CONSUMPTION Cortex Agents → Teams (MCP)

01

Interview

Talk to the people who run the process today.

02

Draft the process

Write it down as steps, with the decisions in between.

03

Infer the data assets

Each step needs something. That something is a data asset.

04

Mark the nexus points

The places where evaluation belongs.

Effective data modeling.

CONTEXT Column metadata (Snowflake)

SEMANTICS Snowflake Semantic Views

CONSUMPTION Cortex Agents → Teams (MCP)

Data modeling splits across three subcategories – **conceptual, logical & physical.**

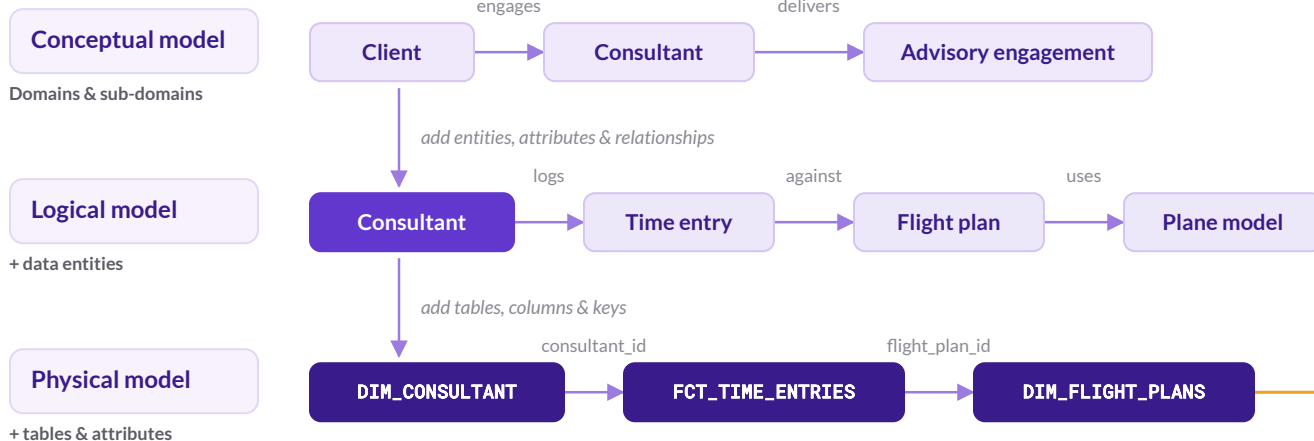
That separation lets different teams work on different components simultaneously.

The physical model converts into tables – and the tables roll up into data products.

MODELS

DATA OWNERSHIP, SECURITY, COMPLIANCE

**PRODUCT OWNERSHIP,
DEFINITIONS, QUALITY**



Concepts only – no keys, no tables.

**FOUNDATIONAL DATA
PRODUCTS**

Consultant Client
Flight Fleet

Metadata.

CONTEXT Column metadata (Snowflake)

SEMANTICS Snowflake Semantic Views

CONSUMPTION Cortex Agents → Teams (MCP)

Metadata provides the context around your artifacts — and it's the jumping-off point for security. One pane of glass.

- ◆ Some is **built by tools** — Snowflake native lineage, the dbt DAG, native column descriptions.
- ◆ Some must be **investigated** — interviews with stakeholders to identify stewards and data domains.
- ◆ Some **surfaces over time** — most-used queries, on a dashboard.

CONTEXT TYPE	EXAMPLE — ST-EXUPÉ CORP
Data domain	"Flight Operations," "Client Advisory," "Fleet"
Steward / owner	Billable-hours metrics owned by Flight Ops; escalation contact: A. Rivière
Lineage	<code>fct_time_entries.billable_hours</code> derives from <code>stg_timesheets.entries</code> , transformed in <code>int_entry_totals</code>
Column metadata	<code>consultant.email</code> — classified PII, masked for roles outside <code>PII_READER</code>
Descriptions	<code>is_active_client</code> : TRUE if a flight plan was filed in the trailing 90 days
Usage metrics	Most-used tables and most popular queries, surfaced over time

Content.

CONTEXT Column metadata (Snowflake)

SEMANTICS Snowflake Semantic Views

CONSUMPTION Cortex Agents → Teams (MCP)

Entities

Logical or conceptual, as defined in context.

Descriptions

Pulled from context, not rewritten here.

Metrics

Defined deterministically.

Relationships

How the entities join.

Golden questions

A baseline you can score future versions against.

CAREFUL WITH THE GOLDEN QUESTIONS

Too many and you get overfitting — plus a high inference cost every time you evaluate.

Content.

CONTEXT Column metadata (Snowflake)

SEMANTICS Snowflake Semantic Views

CONSUMPTION Cortex Agents → Teams (MCP)

IN THIS EXAMPLE

St-Exupé Corp surfaces insights through **Teams**, for their internal analytics team.

Classic BI

You would build visuals.

Agent design

Added step: create the agent — prompt and configuration — and define the tools it can reach.

ON SECURITY

Enforcement *can* happen at this layer. Prioritize it upstream anyway — that's what makes the experience consistent across every stakeholder and every surface.

03

CHANGE MANAGEMENT

Iteration.

How do you change any of this without breaking what already works?

STORY

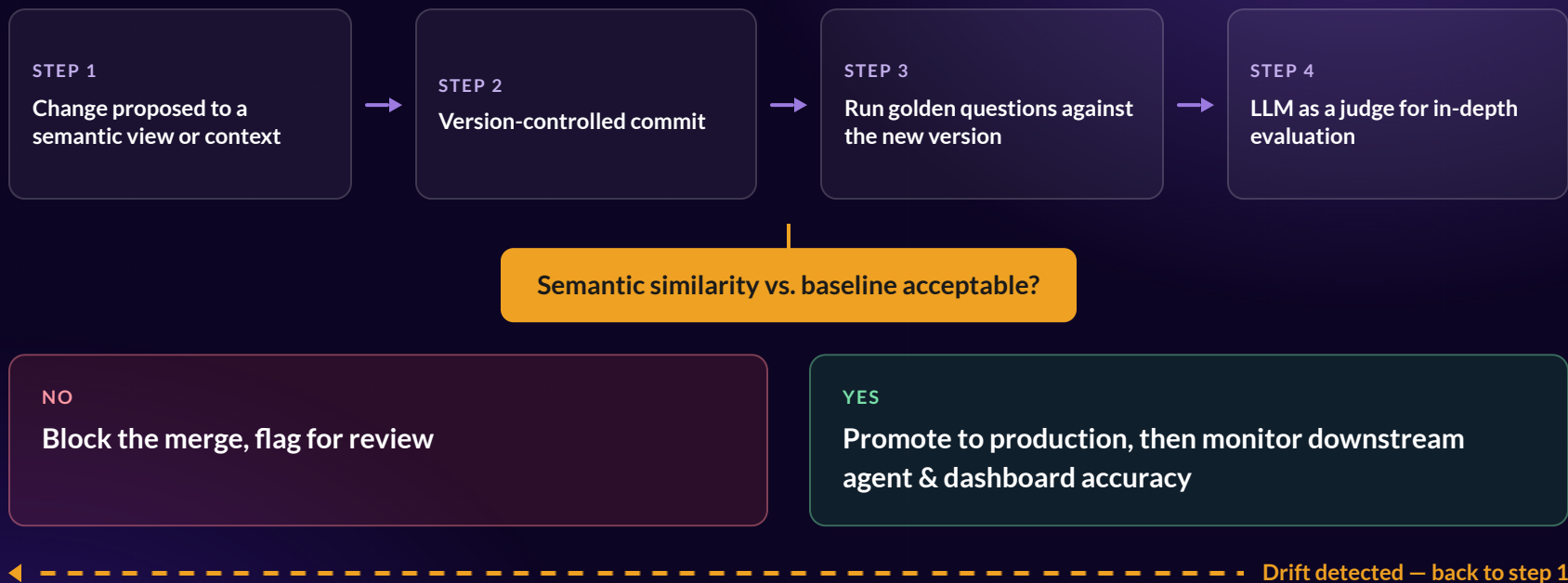
Alright — we've drawn our sheep. But what if we asked for a **new** sheep?

How do we distinguish it from the old one?

How do we evaluate whether it's any better?

THE MECHANISM

Semantic updates.



Syncing all the layers.

SAME REPOSITORY

Automate inside version control

dbt models, Liquibase and friends: changes perpetuate through the repo and trigger CI/CD directly.

SEPARATE REPOSITORIES

Version & contract

Semantic versioning plus a contract — the same discipline you'd apply to an API.

OUR CURRENT BLOCKER: EVALUATION SPEED

- ◆ A single run: inference takes **30 seconds and up**.
- ◆ Volume of changes: context changes constantly, and only some of it is relevant to semantics.

EVALUATE IN PLACE

Agents must be evaluated **within their own environment**.
Recreating an agent "with the same configs" on another platform doesn't replicate the underlying architecture.

04

WRAPPING UP

Conclusion.

Three questions asked at the start. Three answers.

Three questions, answered.

01 Core components: context, semantics, consumption — with security across all three.

Different jobs. Easy to conflate, costly to.

02 How to build it once: model the context, define the semantics, then design the surface.

In that order — each layer feeds the next.

03 How to iterate: golden questions, similarity scoring, gated in the SDLC.

Version-controlled artifacts, not one-off modeling projects.

NEXT STEPS

- ◆ Improving on the existing CI/CD framework.
- ◆ A unified definition of agent design, so changes can be evaluated cross-platform.

CLOSING

Thank you.

Questions, or reach out any time.

REACH OUT

[linkedin.com/in/stéphane-burwash](https://www.linkedin.com/in/stéphane-burwash)

FOLLOW

medium.com/@stefburwash

OUR WEBSITE

[Infostrux.com](https://infostrux.com)

FEEDBACK

Would love to hear your Feedback!

Please let us know your thoughts on the talk! It helps shape the next one, and is very much appreciated!



SCAN TO SHARE FEEDBACK